Why is productive TRA/TRB so frequent in the V4.2 NK boundary?

A four-hypothesis forensic audit of 475,953 cells. This report is diagnostic and does not redefine NK truth or mutate source data.

78.4% raw productive TRA/TRB	**74.3%** of AB calls from join-red-flag sources	**1.5%** one-UMI-only among AB calls with observed UMI support	**1,508** conservative residual AB-compatible cells

Main conclusion. The high TRA/TRB rate cannot be interpreted as an inconvenient biological fact. The dominant detected problem is upstream TCR assignment provenance: legacy barcode-only joins are documented, exact TRA/TRB pairs recur across multiple sample or donor keys in several sources, and some source objects assign TRA/TRB broadly to author-labeled B/myeloid/NK cells. UMI depth cannot repair a receptor copied to the wrong cell.

Ambient reads are a minority explanation in the auditable subset. Only 47 productive-AB cells are supported exclusively by one observed UMI. UMI support is unavailable for 370,292 AB cells, so ambient contamination cannot be ruled out there; it also cannot explain cross-donor reuse of exact paired receptors.

The UMAP did not create the TCR conflict. UMAP is a visualization of the 30-dimensional scVI latent representation. The 4,000-gene model forced core T/NK lineage genes and deliberately excluded TCR V/J/D genes. Clusters retain distinct CD3/T/NK expression gradients, while suspect TRA/TRB assignments span all clusters. Re-clustering alone would not correct an unsafe metadata join.

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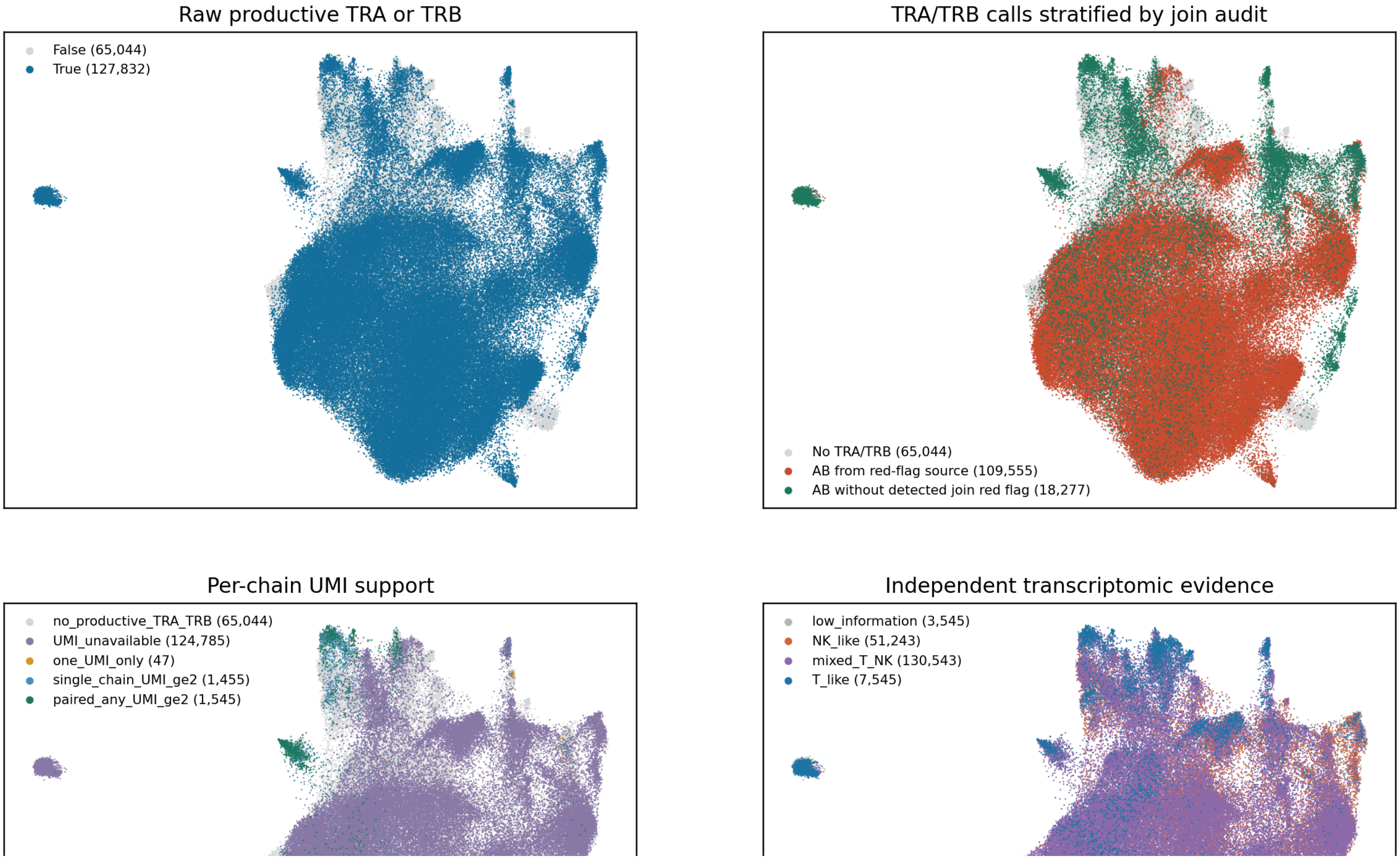
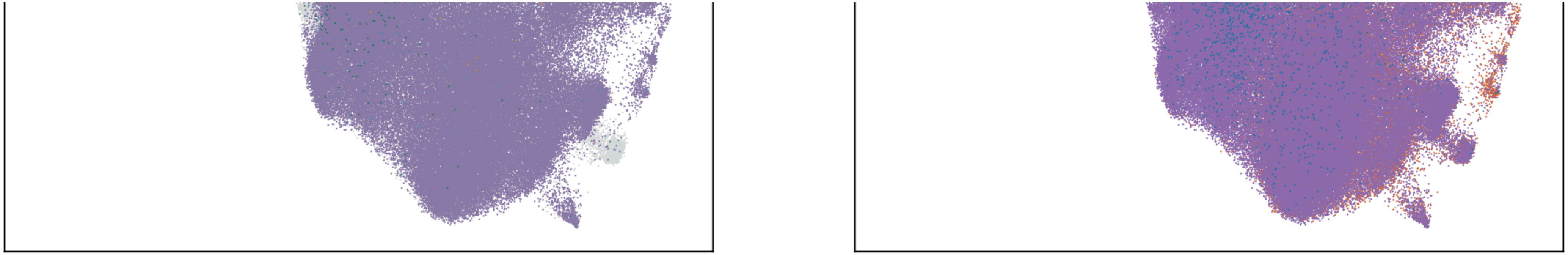
1. Scope and definitions

The boundary is the frozen union of refined parent clusters 9 and 18 from the T/NK-restricted V4.2 integration. A productive chain call means a nonempty productive-filtered CDR3 field. Zero UMI/read values are treated as unavailable because legacy import code used zero when support was absent. A one-UMI-only call requires an observed positive UMI value and no chain with two or more UMIs.

A source receives a join red flag only from auditable evidence: a documented barcode-only mapping mode, extensive exact paired-receptor reuse across sample/donor keys, or strong conflict with source author lineages. This is a quarantine signal, not an estimate of how many individual calls are wrong.

V4.2 boundary: productive TRA/TRB provenance is not a cell-type label





The same cells are colored four ways. Red marks productive alpha/beta calls inherited from a source with a join red flag; expression classes are descriptive multi-gene evidence, not hard truth labels.

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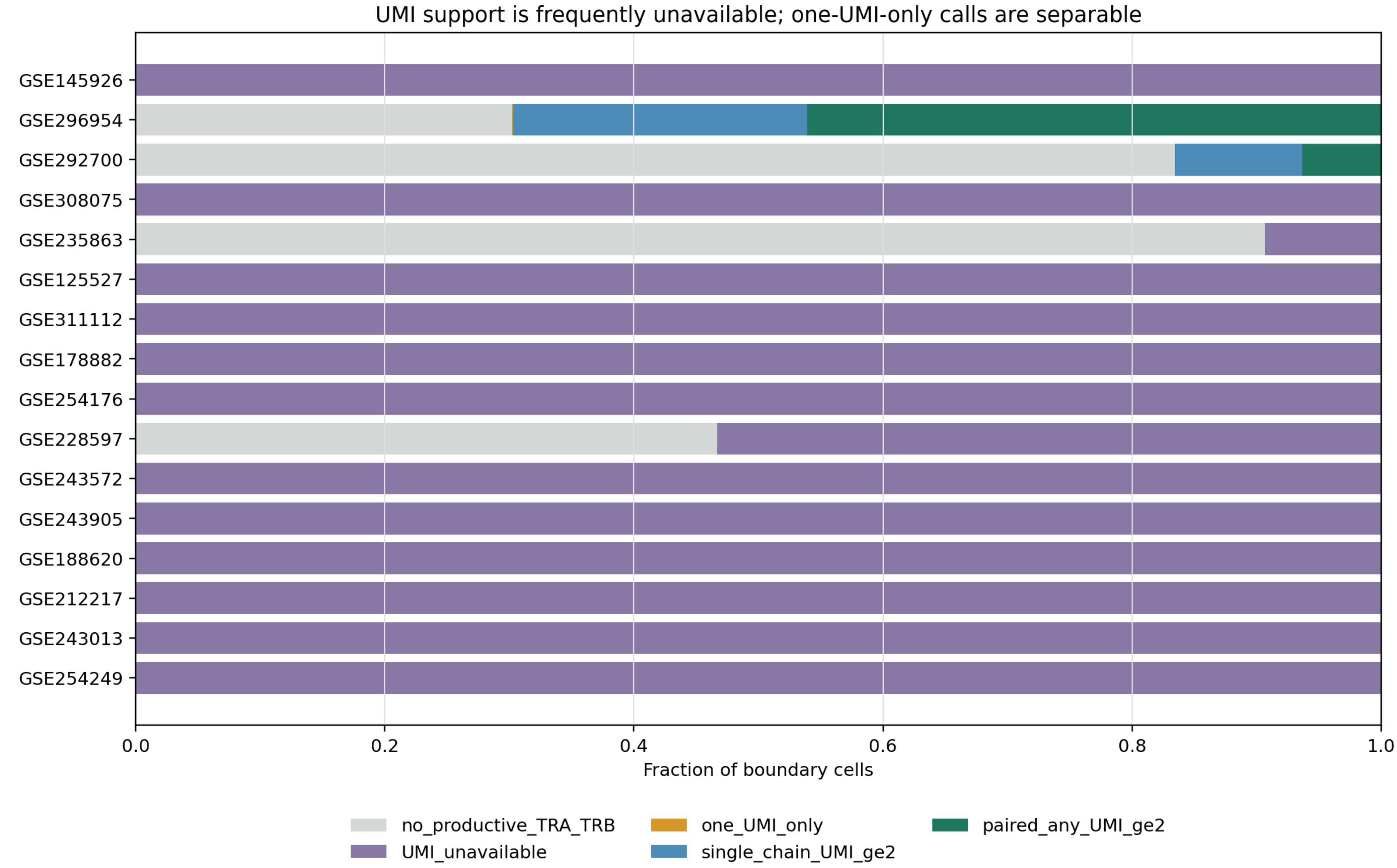
2. Hypothesis 1: ambient TCR reads

Among 3,047 productive-AB cells with at least one observed positive chain UMI, 47 (1.54%) have no chain above one UMI. In contrast, 3,000 have at least one chain with two or more UMIs, and 3,047 have at least ten reads for a chain. This argues against ambient one-UMI calls as the main explanation in sources with quantitative support.

Important limitation: 370,292 productive-AB cells have no usable UMI field. They remain unresolved for ambient contamination. Moreover, high UMI/read support validates the receptor in its originating VDJ library, but not the RNA cell to which it was joined.

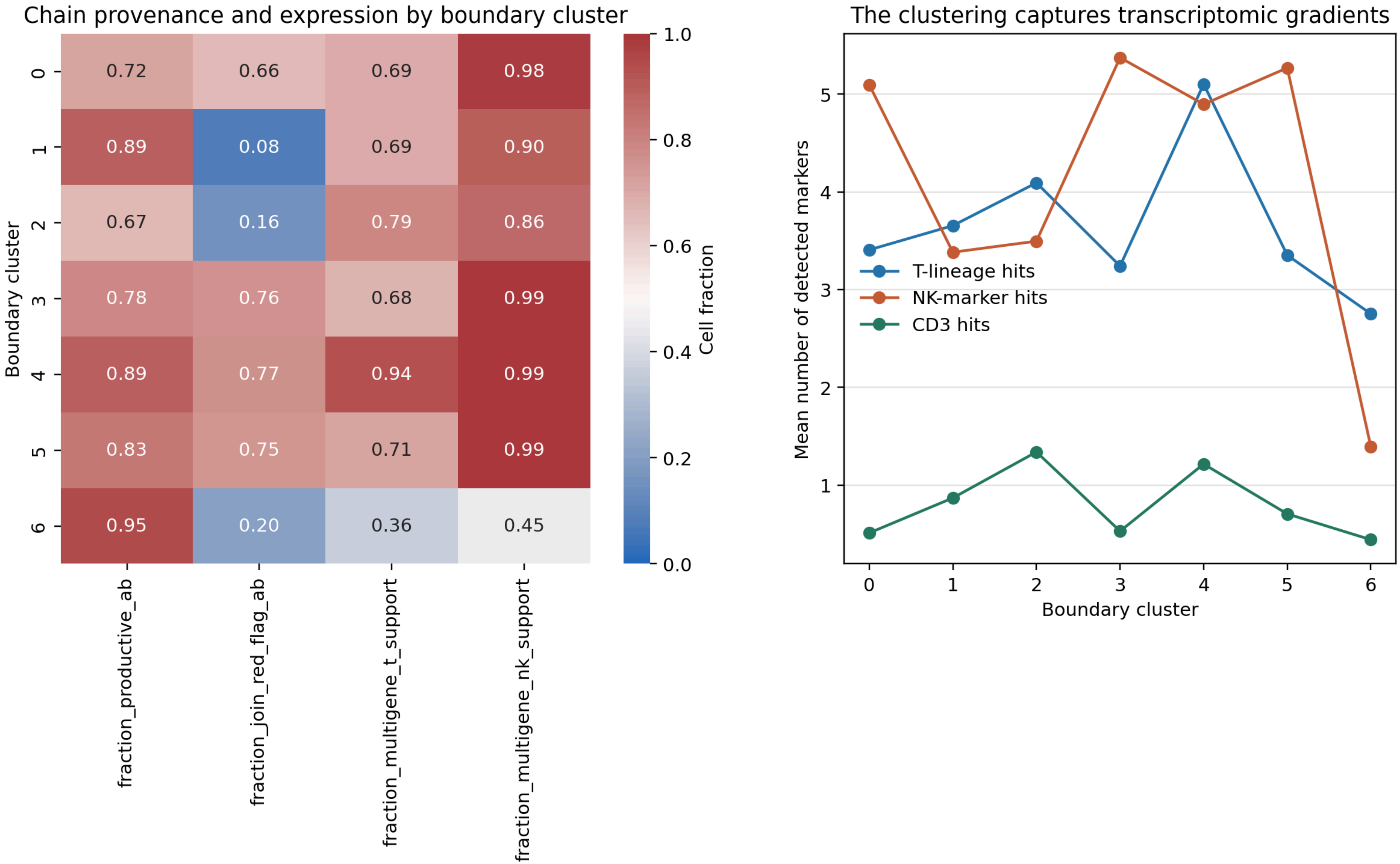


UMI support tiers by source. Purple indicates missing quantitative support, not one UMI.

3. Hypothesis 2: non-ideal HVGs, UMAP, or clustering

The refined scVI model used 4,000 genes. It forced 27/27 specified T/NK lineage genes into the model. TCR V/J/D genes were intentionally excluded so clonotype usage would not drive atlas geometry; constant-region and signaling genes such as `CD3D/E/G`, `TRAC`, `TRBC1/2`, `LCK`, `TRAT1`, `BCL11B`, `KLRD1`, `FCER1G`, and `TYROBP` were retained.

Boundary cluster association is quantified below. The cluster-expression association should be interpreted alongside the much broader source and raw-chain associations. A better boundary-specific integration may sharpen biology after TCR repair, but it cannot make an unsafe receptor join valid.



Clusters show transcriptomic gradients even though raw productive TRA/TRB calls are pervasive. The red-flag component demonstrates why raw chain overlay is not valid NK/T truth.

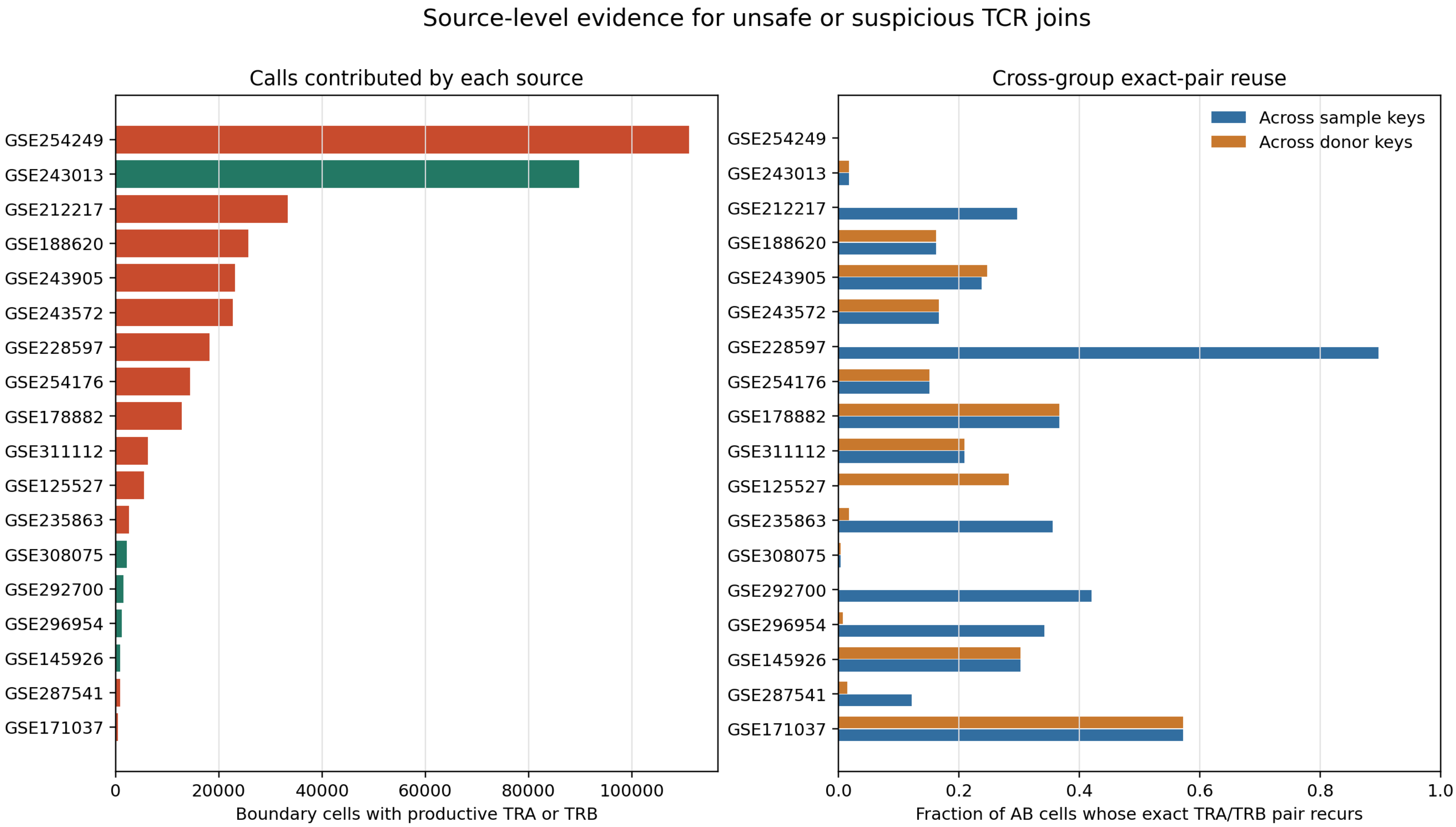
variable	normalized_mutual_information_with_boundary_cluster	cramers_v_with_boundary_cluster
productive_ab	0.02405	0.2195
expression_class	0.07551	0.3087
source_gse_id	0.2147	0.4578
join_audit_status	0.161	0.3671

gene	selected_for_refined_scvi	role
CD3D	True	T_lineage
CD3E	True	T_lineage
CD3G	True	T_lineage
CD247	True	T_lineage
TRAC	True	T_lineage
TRBC1	True	T_lineage
TRBC2	True	T_lineage
TRDC	True	NK_or_context
TRGC1	True	NK_or_context
TRGC2	True	NK_or_context
LCK	True	T_lineage
TRAT1	True	T_lineage
BCL11B	True	T_lineage
THEMIS	True	T_lineage
KLRD1	True	NK_or_context
NCR1	True	NK_or_context
FCGR3A	True	NK_or_context
KLRC1	True	NK_or_context
KLRF1	True	NK_or_context
S1PR5	True	NK_or_context
FCER1G	True	NK_or_context
TYROBP	True	NK_or_context

gene	selected_for_refined_scvi	role
LST1	True	NK_or_context
AIF1	True	NK_or_context
CTSS	True	NK_or_context
LILRB1	True	NK_or_context
LILRB2	True	NK_or_context

4. Hypothesis 3: TCRs joined to the wrong cells

The project SOP states that historical barcode-only or barcode-prefix joins are unsafe because 10x barcodes are reused across samples. The legacy mapping report explicitly records such modes for affected sources. Independently, exact TRA/TRB pairs recur across multiple sample or donor keys, sometimes in a large fraction of assigned cells. Exact paired alpha/beta receptors shared at this scale across unrelated groups are not a plausible ambient-RNA pattern.



Red bars are sources with a documented or strong join red flag. Cross-group pair reuse is calculated from the source H5AD where suitable grouping keys are present.

Source overview

source	boundary_cells	productive_AB	AB_fraction	join_audit	recorded_mapping
GSE254249	111081	111081	1	documented_unsafe_barcode_join	barcode_only
GSE243013	89772	89754	0.9998	no_join_failure_detected	not_recorded
GSE212217	33344	33344	1	highly_suspicious_cross_sample_reuse	not_recorded
GSE188620	25755	25755	1	highly_suspicious_cross_donor_reuse	not_recorded
GSE243905	23166	23166	1	highly_suspicious_cross_donor_reuse	not_recorded
GSE243572	22680	22680	1	highly_suspicious_cross_donor_reuse	not_recorded
GSE228597	34123	18184	0.5329	highly_suspicious_cross_sample_reuse	not_recorded
GSE254176	14373	14372	0.9999	highly_suspicious_cross_donor_reuse	not_recorded
GSE178882	12780	12779	0.9999	highly_suspicious_cross_donor_reuse	not_recorded
GSE311112	6224	6223	0.9998	highly_suspicious_cross_donor_reuse	not_recorded
GSE125527	5517	5514	0.9995	highly_suspicious_cross_donor_reuse	not_recorded
GSE235863	28281	2640	0.09335	highly_suspicious_cross_sample_reuse	not_recorded
GSE308075	2186	2186	1	no_join_failure_detected	not_recorded
GSE292700	9338	1548	0.1658	no_join_failure_detected	not_recorded
GSE296954	1703	1188	0.6976	no_join_failure_detected	not_recorded
GSE145926	865	865	1	no_join_failure_detected	not_recorded
GSE287541	23459	828	0.0353	highly_suspicious_cross_sample_reuse	not_recorded
GSE171037	438	414	0.9452	highly_suspicious_cross_donor_reuse	not_recorded
GSE211504	325	325	1	highly_suspicious_cross_donor_reuse	not_recorded
GSE294273	1322	122	0.09228	source_h5ad_unavailable	not_recorded
GSE287301	3098	108	0.03486	no_join_failure_detected	not_recorded
GSE252762	83	83	1	no_join_failure_detected	not_recorded
GSE159251	142	74	0.5211	no_join_failure_detected	not_recorded
GSE240865	537	72	0.1341	no_join_failure_detected	not_recorded
GSE234069	478	25	0.0523	no_join_failure_detected	not_recorded

source	boundary_cells	productive_AB	AB_fraction	join_audit	recorded_mapping
GSE114724	33	9	0.2727	no_join_failure_detected	not_recorded
GSE162498	318	0	0	no_join_failure_detected	not_recorded
GSE267645	19168	0	0	no_join_failure_detected	not_recorded
GSE301528	5364	0	0	no_join_failure_detected	not_recorded

Exact-pair reuse and lineage conflicts

source	sample_key	AB_exact_pair_cross_sample	donor_key	AB_exact_pair_cross_donor	author_NK_with_AB	author_nonT_with_AB
GSE254249	NaN	0	NaN	0		
GSE243013	sampleID	0.01807	sampleID	0.01807	0.9999	
GSE212217	orig.ident	0.2972	NaN	0		
GSE188620	sample	0.1619	donor	0.1619		
GSE243905	sample	0.2382	donor	0.2469	0.4919	
GSE243572	sample	0.167	donor	0.167		
GSE228597	sample_id	0.8969	NaN	0		
GSE254176	sample	0.1509	donor	0.1509		
GSE178882	sample	0.3673	donor	0.3673		
GSE311112	sample	0.2094	donor	0.2094		
GSE125527	NaN	0	patient_assignment	0.2833	1	1
GSE235863	library_id	0.3555	donor_id	0.01778		0
GSE308075	sample	0.00413	donor	0.00413		
GSE292700	library_id	0.42	donor_id	0.0009358		
GSE296954	library_id	0.342	donor_id	0.007775		
GSE145926	sample	0.3028	donor	0.3028		
GSE287541	library_id	0.1221	donor_id	0.01475	0.04151	
GSE171037	sample	0.5729	donor	0.5729		
GSE211504	sample	0.2758	donor	0.2758		
GSE294273	NaN		NaN			

source	sample_key	AB_exact_pair_cross_sample	donor_key	AB_exact_pair_cross_donor	author_NK_with_AB	author_nonT_with_AB
GSE287301	library_id	0.03884	donor_patient	0.009352		
GSE252762	sample	0.08759	donor	0.08759		
GSE159251	library_id	0.2192	donor_id	0.00305		
GSE240865	library_id	0.07292	NaN	0		
GSE234069	library_id	0.4739	donor_id	0	0.2	
GSE114724	library_id	0.3692	donor_id	0.0002603		
GSE162498	sample	0	donor	0		
GSE267645	sample	0	donor	0		
GSE301528	sample	0	donor	0		

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5. Hypothesis 4: these are truly alpha-beta T cells

Some cells are likely genuine alpha-beta T cells: the boundary was deliberately broad, cytotoxic T and NK transcriptomes overlap, and productive paired receptors with multi-gene T-lineage expression are biologically coherent. However, truth can only be assessed after quarantining red-flag sources.

The conservative residual count is 1,508: paired TRA/TRB, no detected source join red flag, at least one chain with two or more observed UMIs, and multi-gene T-lineage support. A further 66,270 cells are compatible with alpha-beta T identity but lack quantitative UMI support. These are descriptive audit strata, not replacement ground-truth labels.

stratum	n_cells	n_productive_ab	fisher_odds_ratio_multigene_t_support	fisher_p	median_t_hits_ab	median_t_hits_no_ab	mannwhitney_u	mannwhitney_p	auc_t_marker_hits_for_ab
all_boundary	475953	373339	1.584	0	4	3	2.257e+10	0	0.5892
join_red_flag_sources	341546	277305	2.074	0	4	3	1.134e+10	0	0.6368
sources_without_join_red_flag	134407	96034	0.8891	8.611e-18	4	4	1.815e+09	1.109e-05	0.4924

6. Decision and corrective action

1. Quarantine productive TRA/TRB fields from every documented or strongly suspicious source for model truth and NK exclusion. Do not delete the original fields; add provenance-qualified replacements.
2. Rebuild affected datasets from raw productive VDJ contigs using the canonical `sample_id + barcode_core` key, retaining UMI/read support and stopping when sample identity is ambiguous.
3. Validate each rebuilt source by key uniqueness, before/after coverage, exact receptor reuse across donors, and author-lineage conflicts. A high UMI count alone is not a join validation.
4. Propagate only validated receptor fields into harmonized metadata, then regenerate the T/NK subset, HVGs, scVI latent space, and boundary clustering.
5. Keep TCR V/J/D genes out of integration HVGs unless performing a separate sensitivity analysis; use them as classifier features or overlays after join repair, not as drivers of cell-state geometry.

Generated by `workflows/gdtai/audit_gdtai_v4_2_boundary_tcr_conflicts.py`. Source H5AD files were read only.